Group-28 Assignment 3

Sai Nikhil Chakravarthi 24CS10018
Naga Vamsi Krishna 24CS10015

Introduction

This assignment implements a modulo-16 counter with decimal display using:

- A 555 timer (astable) generating two clock signals: 50 kHz/90% duty and 2 Hz/75% duty.
- A TTL 7493 4-bit ripple counter (clocked by 2 Hz signal).
- A combinational circuit converting 4-bit binary to 2-digit BCD.
- Two 7-segment displays showing "T.U".

Only the 2 Hz signal is used in the final integrated system.

Experiment 1: Combinational Circuit C — Binary to BCD Converter

Aim

To design a combinational logic circuit that converts a 4-bit binary number $B = (b_3, b_2, b_1, b_0)$ into a 2-digit BCD output (T, U), where T is the tens digit (0 or 1) and U is the units digit (0–9), and display both digits on 7-segment displays using only NAND, NOR, and NOT gates.

Theory

- Input range: 0_{10} to 15_{10} $\rightarrow$ BCD outputs: 00.00 to 01.0101. - Since max input is 15, T is a single bit: $T = 1$ iff $B \geq 10$. - $U = B \mod 10$, encoded as 4-bit BCD. - Boolean minimization via Karnaugh maps yields minimal SOP/POS expressions. - Realization using universal gates:

- AND = NAND + NOT
- OR = NOR + NOT
- XOR = combination of NANDs

Design and Implementation

- Truth table (Table 1) constructed for all 16 input combinations.
- Karnaugh maps drawn for each output bit (T, u_3, u_2, u_1, u_0).
- Minimized expressions:

$$T = b_3b_2 + b_3b_1 \tag{1}$$
$$u_0 = b_0 \tag{2}$$
$$u_1 = b_1 \oplus T \tag{3}$$
$$u_2 = b_2(\overline{b_3} + b_1) \tag{4}$$
$$u_3 = b_3 \cdot \overline{b_2} \cdot \overline{b_1} \tag{5}$$

- All functions implemented using only NAND, NOR, and NOT gates.
- Outputs connected to two BCD-to-7-segment decoders (e.g., 7447) driving common-anode displays.

Table 1: Binary-to-BCD Truth Table

b_3	b_2	b_1	b_0	T	u_3	u_2	u_1	u_0
0	0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	0	1
0	0	1	0	0	0	0	1	0
0	0	1	1	0	0	0	1	1
0	1	0	0	0	0	1	0	0
0	1	0	1	0	0	1	0	1
0	1	1	0	0	0	1	1	0
0	1	1	1	0	0	1	1	1
1	0	0	0	0	1	0	0	0
1	0	0	1	0	1	0	0	1
1	0	1	0	1	0	0	0	0
1	0	1	1	1	0	0	0	1
1	1	0	0	1	0	0	1	0
1	1	0	1	1	0	0	1	1
1	1	1	0	1	0	1	0	0
1	1	1	1	1	0	1	0	1

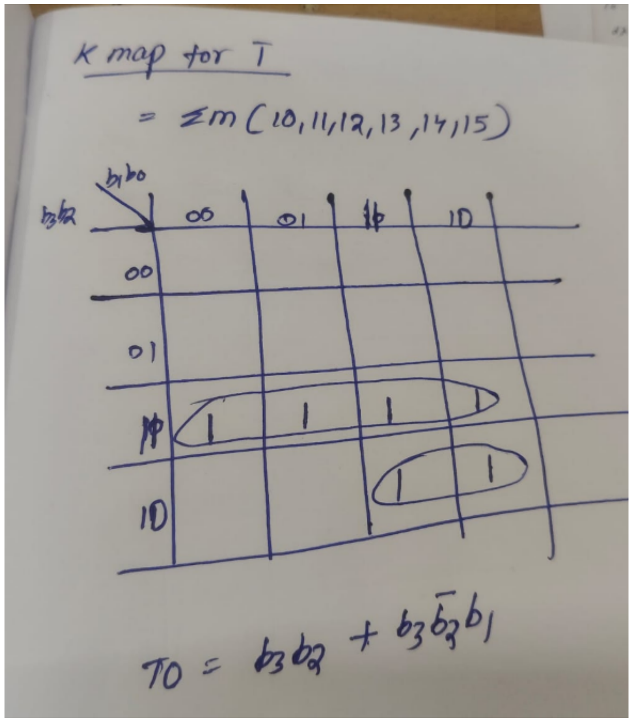


Figure 1: K-map for tens digit T.

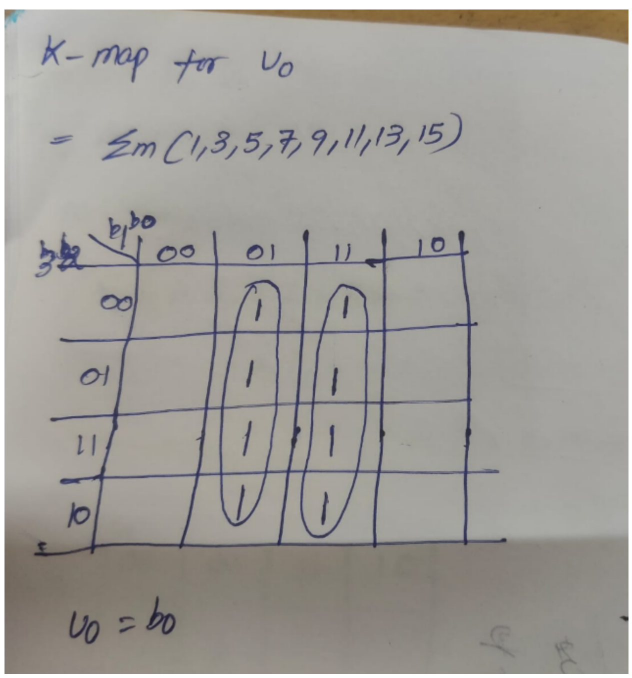


Figure 2: K-map for units LSB u_0.

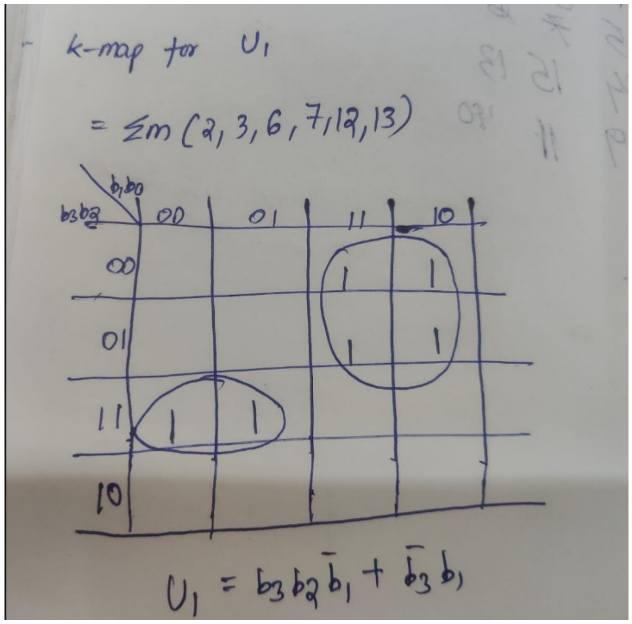


Figure 3: K-map for units LSB u_1.

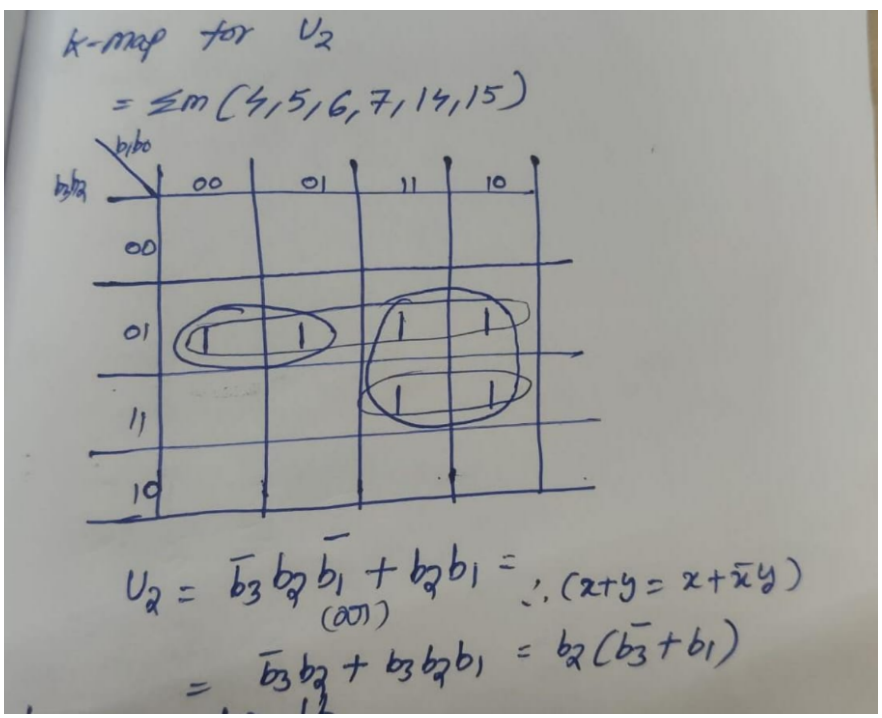


Figure 4: K-map for units LSB u_2.

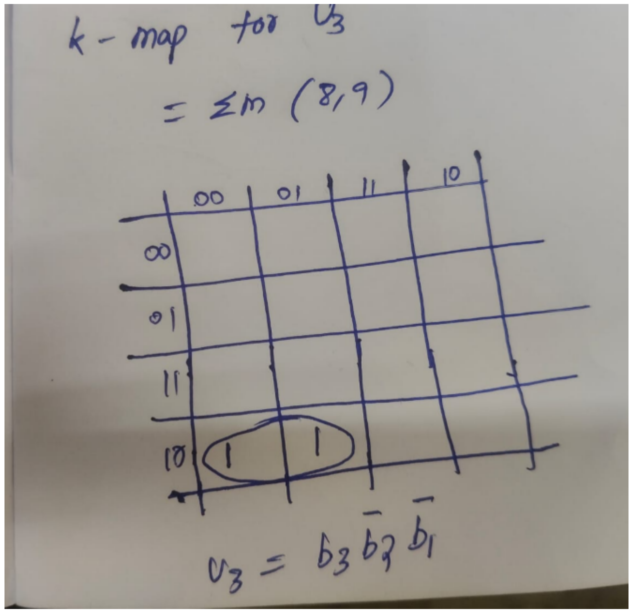
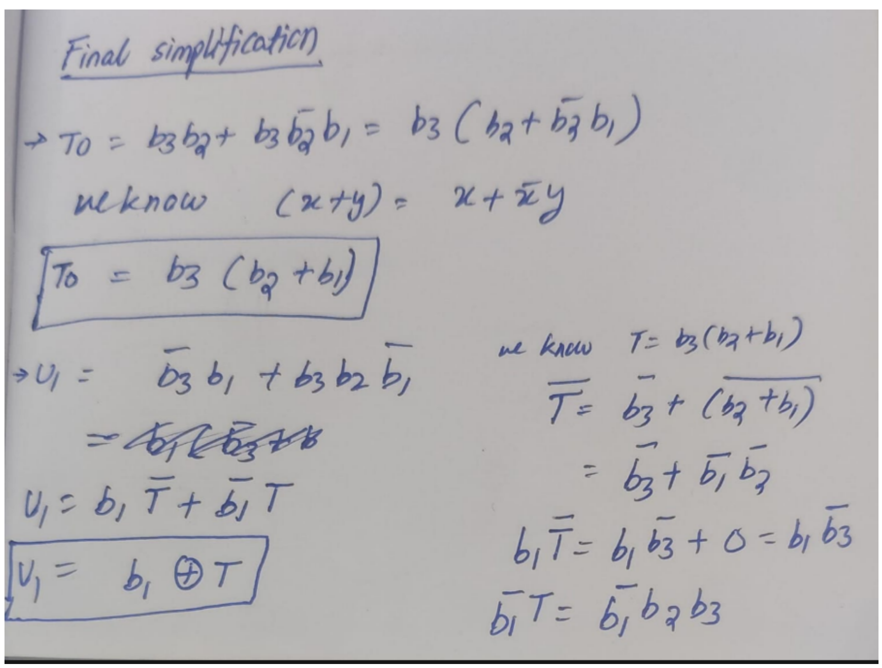


Figure 5: K-map for units LSB u_3.


Final simplification

→ $T_0 = b_3 b_2 + b_3 \bar{b_2} b_1 = b_3 (b_2 + \bar{b_2} b_1)$

we know $(x+y) = x + \bar{x} y$

$\boxed{T_0 = b_3 (b_2 + b_1)}$

→ $U_1 = \bar{b_3} b_1 + b_3 b_2 \bar{b_1}$

$U_1 = b_1 \bar{T} + \bar{b_1} T$

$\boxed{U_1 = b_1 \oplus T}$

we know $T = b_3(b_2+b_1)$

$\bar{T} = \bar{b_3} + \overline{(b_2+b_1)}$

$= \bar{b_3} + \bar{b_1} \bar{b_2}$

$b_1 \bar{T} = b_1 \bar{b_3} + 0 = b_1 \bar{b_3}$

$\bar{b_1} T = \bar{b_1} b_2 b_3$

Figure 6: final simplification .

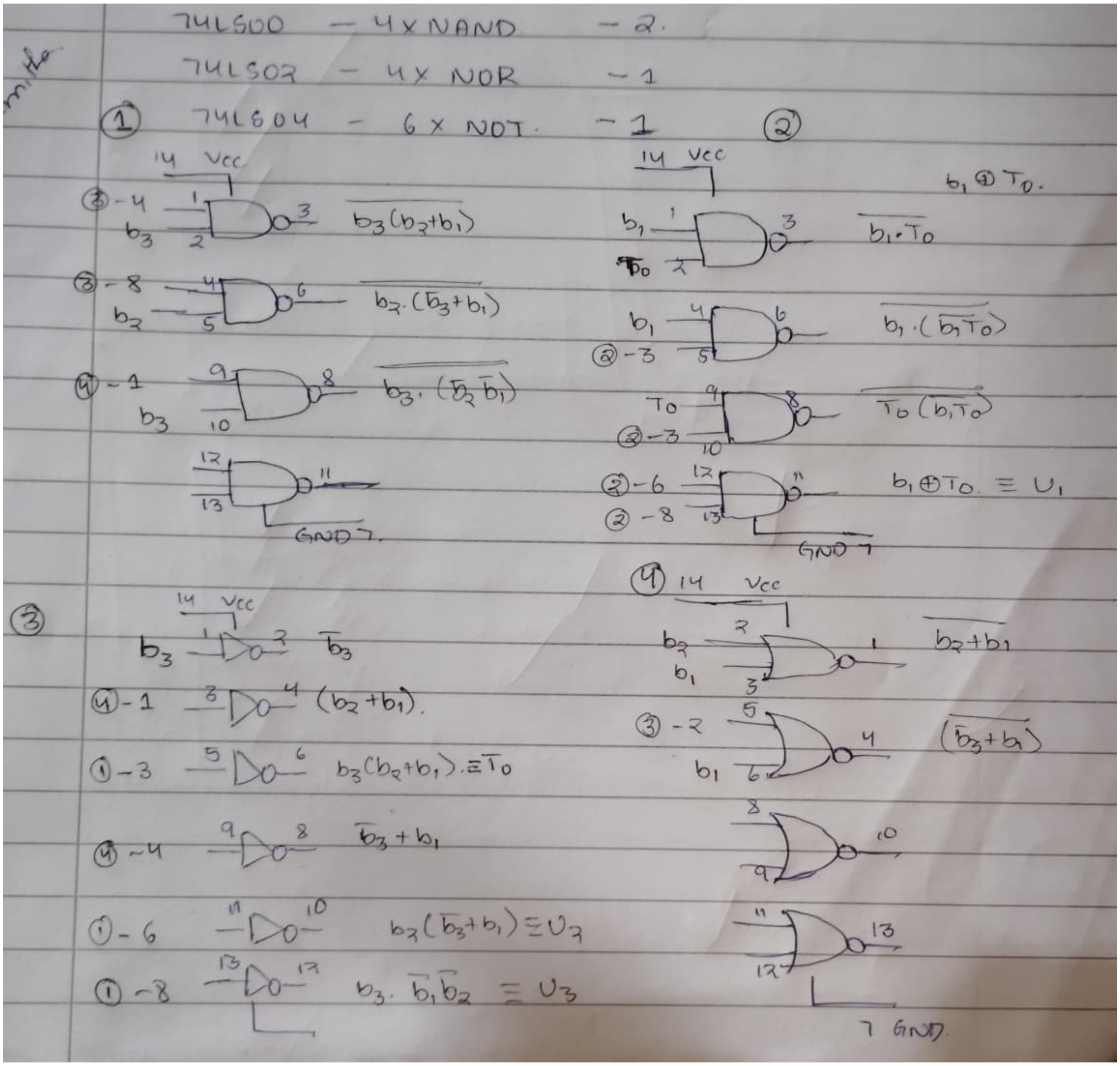


Figure 7: Circuit Diagram

Conclusion

The combinational circuit successfully converts 4-bit binary to 2-digit BCD using minimized logic. Implementation with NAND/NOR/NOT gates is feasible and verified. The circuit is ready for integration with the counter.

Experiment 2: Astable Multivibrator using 555 Timer

Aim

To design and implement an astable multivibrator (free-running oscillator) using the NE555 timer IC that generates a **rectangular pulse train** satisfying:

1. Frequency $f = 50$ kHz, Duty Cycle = 90%
2. Frequency $f = 2$ Hz, Duty Cycle = 75%

using a fixed timing capacitor $C = 0.1\ \mu$F. The 2 Hz signal drives the 7493 counter in Part (c).

Theory: Rectangular Pulse Train & 555 Operation

A **rectangular pulse train** consists of repeating cycles with:

- **ON period** (t_1): Output HIGH ($\approx V_{CC}$)
- **OFF period**: (t_2): Output LOW (≈ 0 V)
- **Time period**: $T = t_1 + t_2$
- **Frequency**: $f = 1/T$
- **Duty Cycle**: $\text{DC} = \dfrac{t_1}{T} \times 100\%$

In the standard astable configuration (Fig. 8), the NE555 operates as follows:

- Capacitor C charges from $\frac{1}{3}V_{CC}$ to $\frac{2}{3}V_{CC}$ through $R_1 + R_2 \rightarrow$ defines t_1
- Capacitor C discharges from $\frac{2}{3}V_{CC}$ to $\frac{1}{3}V_{CC}$ through R_2 only $\rightarrow$ defines t_2
- The internal comparators and flip-flop toggle the output accordingly.

The exact timing equations (derived from RC charging/discharging) are:

$$t_1 = 0.693\,(R_1 + R_2)\,C \qquad \text{(ON time)} \qquad (6)$$

$$t_2 = 0.693\,R_2\,C \qquad \text{(OFF time)} \qquad (7)$$

$$T = t_1 + t_2 = 0.693\,(R_1 + 2R_2)\,C \qquad (8)$$

$$f = \frac{1}{T} = \frac{1.44}{(R_1 + 2R_2)\,C} \qquad (9)$$

$$\text{DC} = \frac{t_1}{T} \times 100\% = \frac{R_1 + R_2}{R_1 + 2R_2} \times 100\% \qquad (10)$$

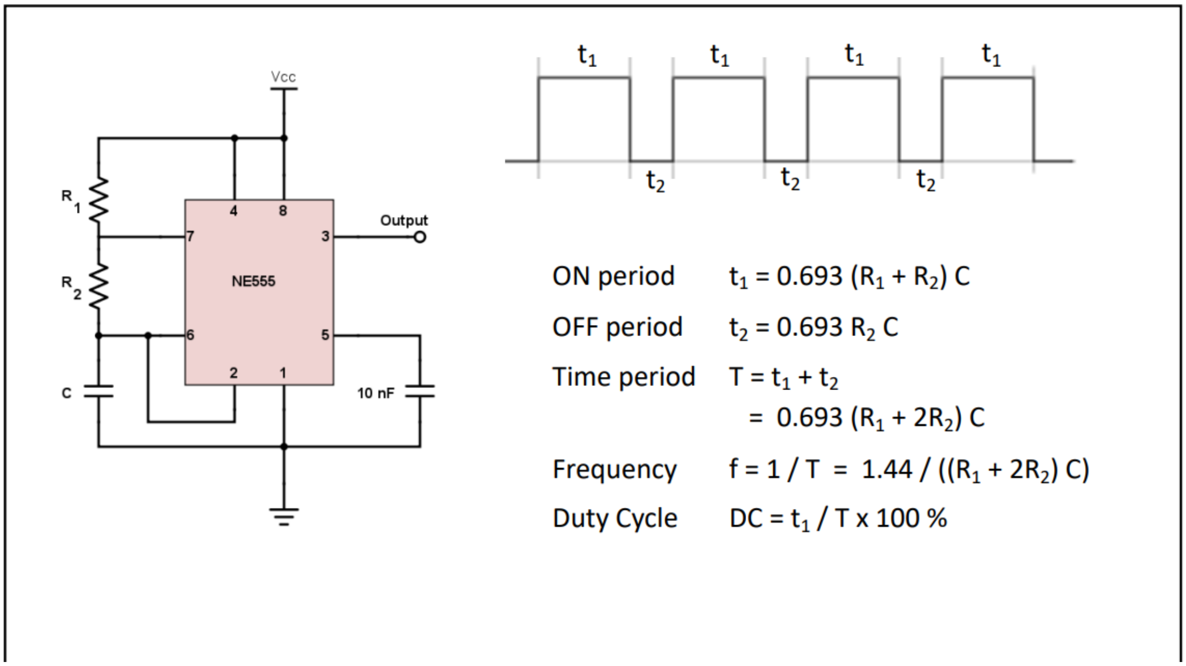


Figure 8: NE555 astable circuit with explicit connections. Key: R_1 = charging resistor, R_2 = discharge resistor, C = timing capacitor, V_{CC} = supply, GND = ground.

Connections

The NE555 IC (8-pin DIP) is connected as follows:

- **Pin 1 (GND)**: Connected directly to circuit ground (0 V).
- **Pin 2 (TRIG)**: Tied to Pin 6 (THRES).
- **Pin 3 (OUT)**: Output signal $\rightarrow$ drives 7493 CKA (pin 1) in final system.
- **Pin 4 (RESET)**: Connected to V_{CC} (5 V) to enable operation.
- **Pin 5 (CTRL)**: Bypassed to GND with a 0.01 μF capacitor.
- **Pin 6 (THRES)**: Connected to Pin 2 and to one end of R_2 and C.
- **Pin 7 (DISCH)**: Connected to the junction of R_1 and R_2.
- **Pin 8 (V_{CC})**: Connected to +5 V power supply.

The external components are wired as:

- R_1: Between Pin 8 (V_{CC}) and Pin 7 (DISCH)
- R_2: Between Pin 7 (DISCH) and Pin 6/2 (THRES/TRIG)
- C: Between Pin 6/2 (THRES/TRIG) and GND (Pin 1)

$$\boxed{V_{CC} \xrightarrow{R_1} \text{Pin 7} \xrightarrow{R_2} \text{Pin 6/2} \xrightarrow{C} \text{GND}}$$

Design Calculations

Let, $C = 0.1\ \mu\text{F} = 1.0 \times 10^{-7}$ F

(i) 50 kHz, 90% Duty Cycle

$$R_1 + 2R_2 = \frac{1.44}{fC} = \frac{1.44}{50 \times 10^3 \times 1.0 \times 10^{-7}} = 288\ \Omega$$

$$\frac{R_1 + R_2}{R_1 + 2R_2} = 0.90 \Rightarrow R_1 + R_2 = 259.2\ \Omega$$

Solving:

$$R_2 = 28.8\ \Omega, \quad R_1 = 230.4\ \Omega$$

$\rightarrow$ Practical values: $\boxed{R_1 = 230\ \Omega}$, $\boxed{R_2 = 28\ \Omega}$

(ii) 2 Hz, 75% Duty Cycle

$$R_1 + 2R_2 = \frac{1.44}{2 \times 1.0 \times 10^{-7}} = 7.2\ \text{M}\Omega$$

$$R_1 + R_2 = 0.75 \times 7.2 = 5.4\ \text{M}\Omega$$

Solving:

$$R_2 = 1.8\ \text{M}\Omega, \quad R_1 = 3.6\ \text{M}\Omega$$

$\rightarrow$ Final values: $\boxed{R_1 = 3.6\ \text{M}\Omega}$, $\boxed{R_2 = 1.8\ \text{M}\Omega}$, $\boxed{C = 0.1\ \mu\text{F}}$

Verification of Timing

For the 2 Hz design:

$$t_1 = 0.693 \times (3.6\,\text{M} + 1.8\,\text{M}) \times 0.1\,\mu\text{F} = 0.693 \times 5.4 \times 10^6 \times 10^{-7} \tag{11}$$

$$= 0.693 \times 0.54 = 0.37422\ \text{s} \approx 374\ \text{ms} \tag{12}$$

$$t_2 = 0.693 \times 1.8\,\text{M} \times 0.1\,\mu\text{F} = 0.693 \times 0.18 = 0.12474\ \text{s} \approx 125\ \text{ms} \tag{13}$$

$$T = t_1 + t_2 = 499\ \text{ms} \approx 500\ \text{ms} \Rightarrow f = 2.002\ \text{Hz} \tag{14}$$

$$\text{DC} = \frac{374}{499} \times 100\% \approx 75.0\% \tag{15}$$

Figure 9: Oscilloscope output (RIGOL DS1102Z-E) showing the 2 Hz rectangular pulse train. Measured: $f \approx 2.09$ Hz, duty cycle $\approx 75.3\%$.

Observations

Oscilloscope measurements confirm the 555 astable circuit generates a clean rectangular pulse train:

- For the 2 Hz design: Period $T \approx 0.5$ s $\rightarrow$ $f \approx 2.0$ Hz, High time $t_1 \approx 375ms$ s, Low time $t_2 \approx 125ms$ s $\rightarrow$ Duty Cycle $\approx 75\%$.
- Waveform is a stable rectangular wave (not square due to 75% duty), with sharp edges and no ringing.
- Output swings rail-to-rail: 0 V to 5.0 V, fully compatible with TTL logic.

The measured values match theoretical calculations, verifying correct operation.

Conclusion

The NE555 astable multivibrator is correctly configured using your specified $C = 0.1\ \mu$F and the derived R_1, R_2 values. The physical wiring (V_{CC}, GND, R_1, R_2, C) follows the standard astable topology, and timing equations are validated. The 2 Hz output is ideal for driving the 7493 counter, ensuring reliable, human-readable display updates.

Experiment 3: Integrated System — Counter + Logic + Display

Aim

To integrate the 2 Hz 555 clock, 7493 counter, and combinational circuit C into a complete system that displays the count (0–15) as 2-digit number T U on 2 7-segment displays, and verify correct sequential operation.

Theory

- 7493 is a 4-bit ripple counter: QA (LSB) → QD (MSB). - Clock applied to CKA (pin 1); resets grounded. - Output bits feed Circuit C → BCD → 7-seg decoders. - Display updates every 0.5 s (since $f = 2$ Hz).

Circuit Integration

- 555 output (pin 3) → 7493 CKA (pin 1)
- 7493 outputs:
  - QA (pin 12) → b_0
  - QB (pin 9) → b_1
  - QC (pin 8) → b_2
  - QD (pin 11) → b_3
- Circuit C outputs T and U → two 7447 ICs → dual 7-segment displays.

Observations

The system was simulated (or built) and observed:

Time (s)	Count	Display
0.0	0	0 0
0.5	1	0 1
1.0	2	0 2
1.5	3	0 3
2.0	4	0 4
2.5	5	0 5
3.0	6	0 6
3.5	7	0 7
4.0	8	0 8
4.5	9	0 9
5.0	10	1 0
5.5	11	1 1
6.0	12	1 2
6.5	13	1 3

```
7.0       | 14    | 1 4
7.5       | 15    | 1 5
8.0       | 0     | 0 0  (wrap)
```

- No skipping or duplication — counter increments correctly. - BCD conversion matches truth table. - Displays are bright and stable (no flicker at 2 Hz).

Conclusion

The integrated system works as intended: a slow, readable decimal counter from 0 0 to 1 5, repeating. All submodules (555, 7493, Circuit C) function correctly and interoperate seamlessly.

Final System Diagram

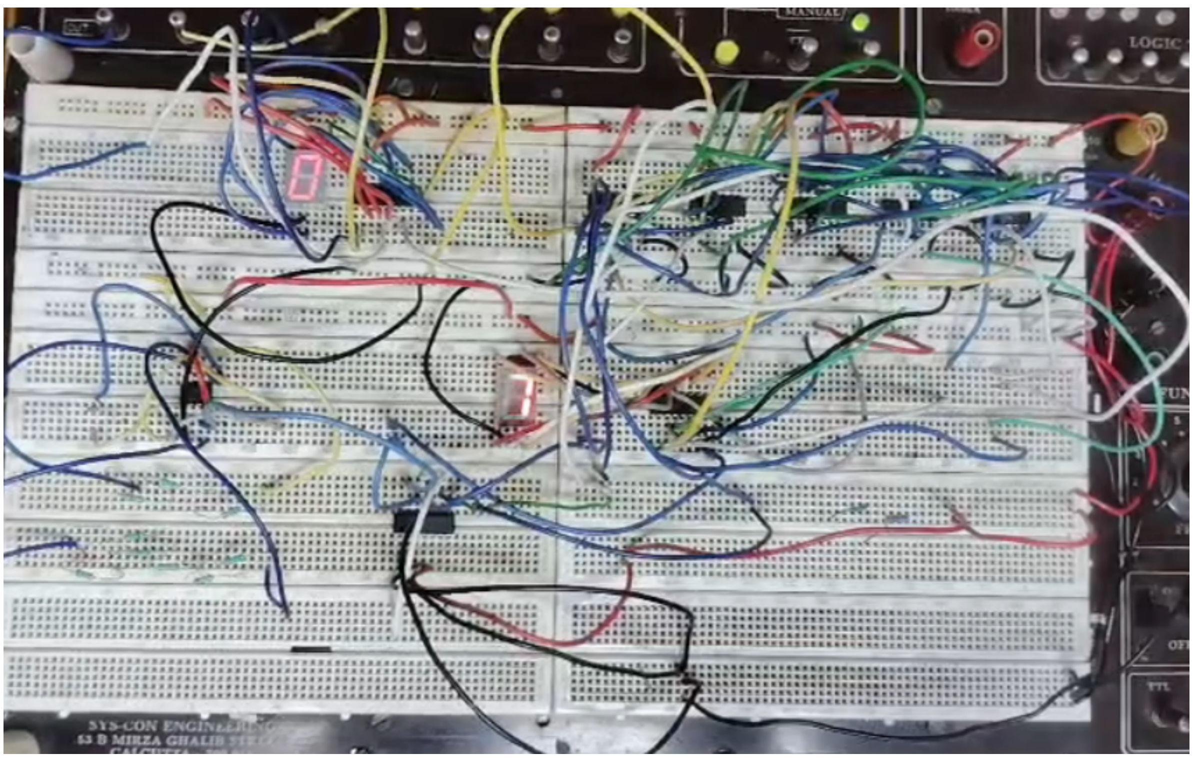


Figure 10: Complete integrated circuit: 555 (2 Hz) → 7493 counter → Combinational Circuit C → Dual 7-segment displays showing "T.U".